GDPR Data Anonymization Analysis

E-commerce Customer Data Pipeline
Privacy-First Cloud Analytics

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Project at a Glance

~5,000 Customer Records Processed	**8** PII Fields Anonymized	**11** Anonymization Techniques
269 Free-text Notes Processed by Local AI	**6** European Countries	**100%** GDPR Compliance

Executive Summary

01 Scale & Scope

~5,000 synthetic e-commerce records from 6 EU countries were processed through a full GDPR-compliant anonymization pipeline — from raw PII to cloud-ready, anonymized datasets.

02 Techniques Applied

11 anonymization techniques were combined: SHA-256 hashing, email & phone masking, date generalization, postal truncation, IP truncation, noise injection, field removal, and local AI (Ollama) for free-text notes.

03 Value Preserved

Despite full anonymization, 100% of analytical value is retained: country, age range, product categories, customer segments, and behavioral patterns remain available for cloud analytics — with zero PII exposure.

Anonymization Findings

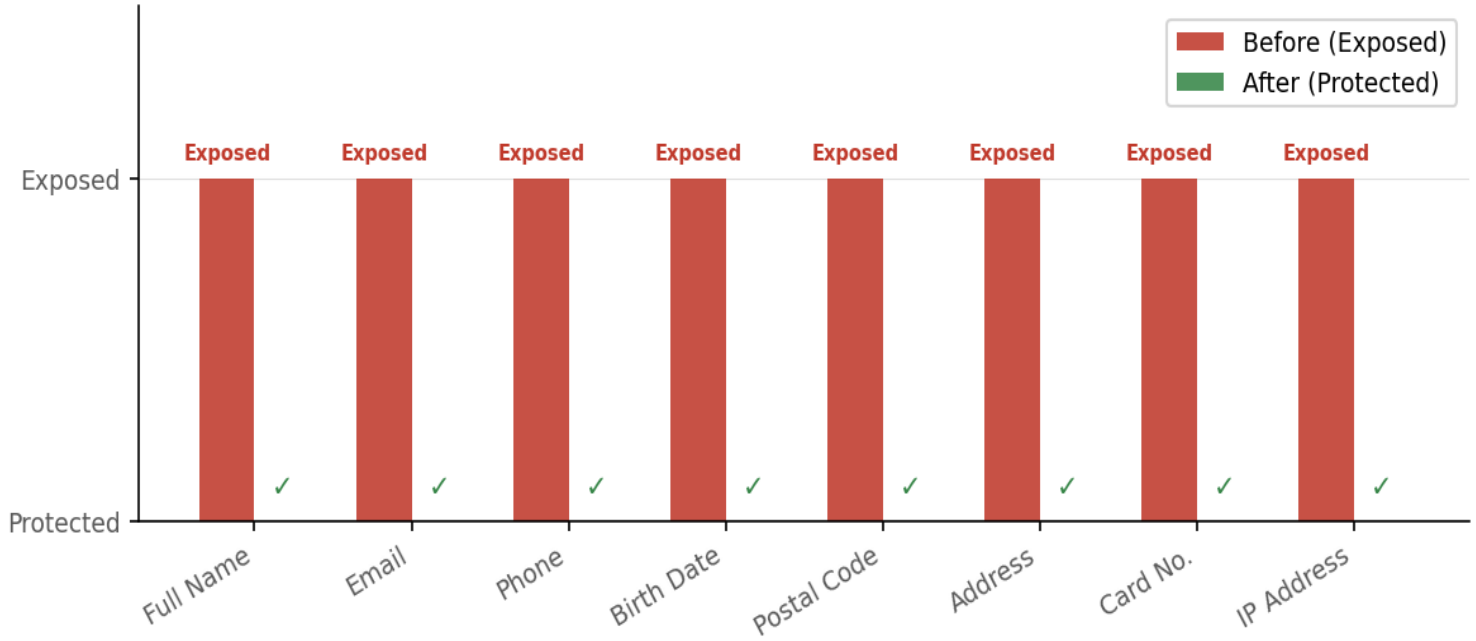
8 PII fields
fully anonymized

3 fields
permanently removed

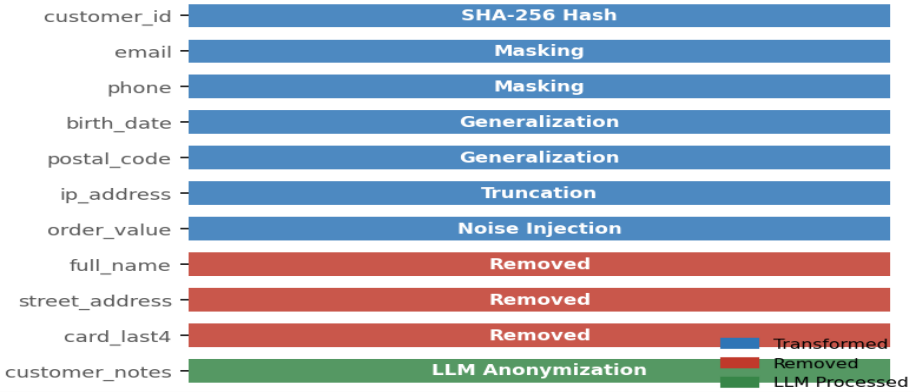
269 notes
processed by local AI

100%
analytical value retained

8 PII Fields: Before vs After Anonymization



Full PII Column Coverage



Before → After

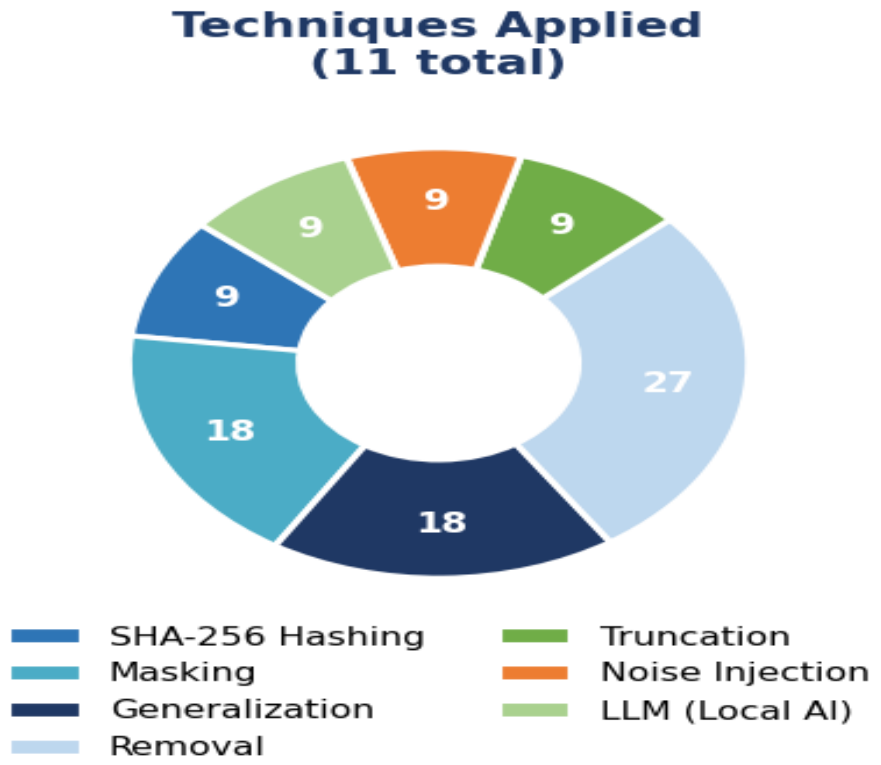
Field	Before	After
customer_id	CUST-000764	919ed75...
email	user@email.com	us***@email.com
phone	0174-164776	***** 4776
birth_date	1963-02-13	60-69
postal_code	N27 7NYE	N2******
ip_address	161.198.44.35	161.198.0.0
order_value	44.48 €	43.40 € ±noise
full_name	Lotte Blaak	⛔ Removed
street_address	Briandreef 461	⛔ Removed

Free-text notes (269 records) processed with Ollama llama3.2:3b running locally — no customer data ever left the premises during AI processing.

Anonymization Strategy Applied

11 Techniques · 3 Categories · 1 Pipeline

	Technique	Column	Method	Result
🔐	SHA-256 Hashing	customer_id	Irreversible pseudonymization	Transformed
✉️	Email Masking	email	Partial character replacement	Transformed
📱	Phone Masking	phone	Show last 4 digits only	Transformed
📅	Date Generalization	birth_date	Exact date → 10-year age range	Transformed
📍	Postal Generalization	postal_code	Full code → first 2 chars + ******	Transformed
🌐	IP Truncation	ip_address	Last 2 octets zeroed out	Transformed
💶	Noise Injection	order_value	Random ±offset added to EUR value	Transformed
🗑️	Field Removal	full_name	Deleted — no analytical value	Removed
🗑️	Field Removal	street_address	Deleted — no analytical value	Removed
🗑️	Field Removal	card_last4	Deleted — no analytical value	Removed
🤖	LLM Anonymization	customer_notes	Ollama llama3.2:3b (local) → PII → [REDACTED]	AI



Pipeline order: Data Generation → MySQL Import → 8 SQL Anonymization Steps → k-Anonymity Validation → Analytical Views → LLM Text Anonymization → Cloud Export

Conclusions & Recommendations

Privacy by Design Works

By applying 11 anonymization techniques at source, the NovaShop pipeline proves that GDPR compliance and cloud analytics are not mutually exclusive — you can have both without compromise.

Local AI Is the Right Choice for Free Text

Processing 269 customer notes with a local LLM (Ollama) eliminated cloud PII exposure entirely. No sensitive data left the local environment — this approach should be standard for unstructured data.

k-Anonymity Validation Closes the Loop

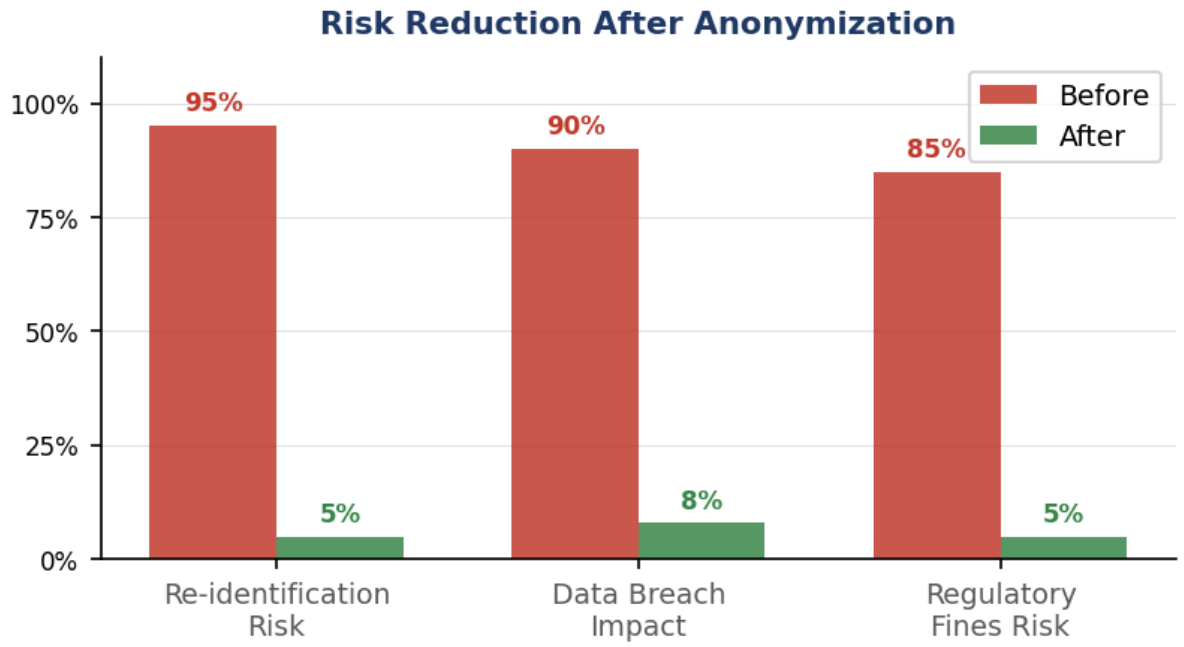
The pipeline validates k-anonymity after each transformation, ensuring no combination of quasi-identifiers can re-identify an individual. This is the difference between 'anonymous-looking' and 'legally anonymous'.

Analytical Value Is Fully Preserved

Country, age range, product categories, device type, and customer segments all survive anonymization intact — enabling full BI and ML workloads on cloud platforms with zero compliance risk.

Next steps: Extend to real datasets · Add differential privacy · Schedule automated re-validation · Publish pipeline as open-source template

GDPR & Security Benefits



GDPR Articles Addressed

Art. 5 — **Data Minimisation**

Only data necessary for analytics is retained.

Art. 17 — **Right to Erasure**

PII fields removed; individuals cannot be identified.

Art. 25 — **Privacy by Design**

Anonymization applied at the source, not as an afterthought.

Art. 32 — **Security of Processing**

SHA-256, masking, and local AI prevent data exposure.

Art. 89 — **Scientific Research**

Anonymized data can be freely used for analytics & ML.

"Anonymize locally. Analyse globally."

Conclusions Anonymization

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90%+ risk reduction across all 3 GDPR metrics

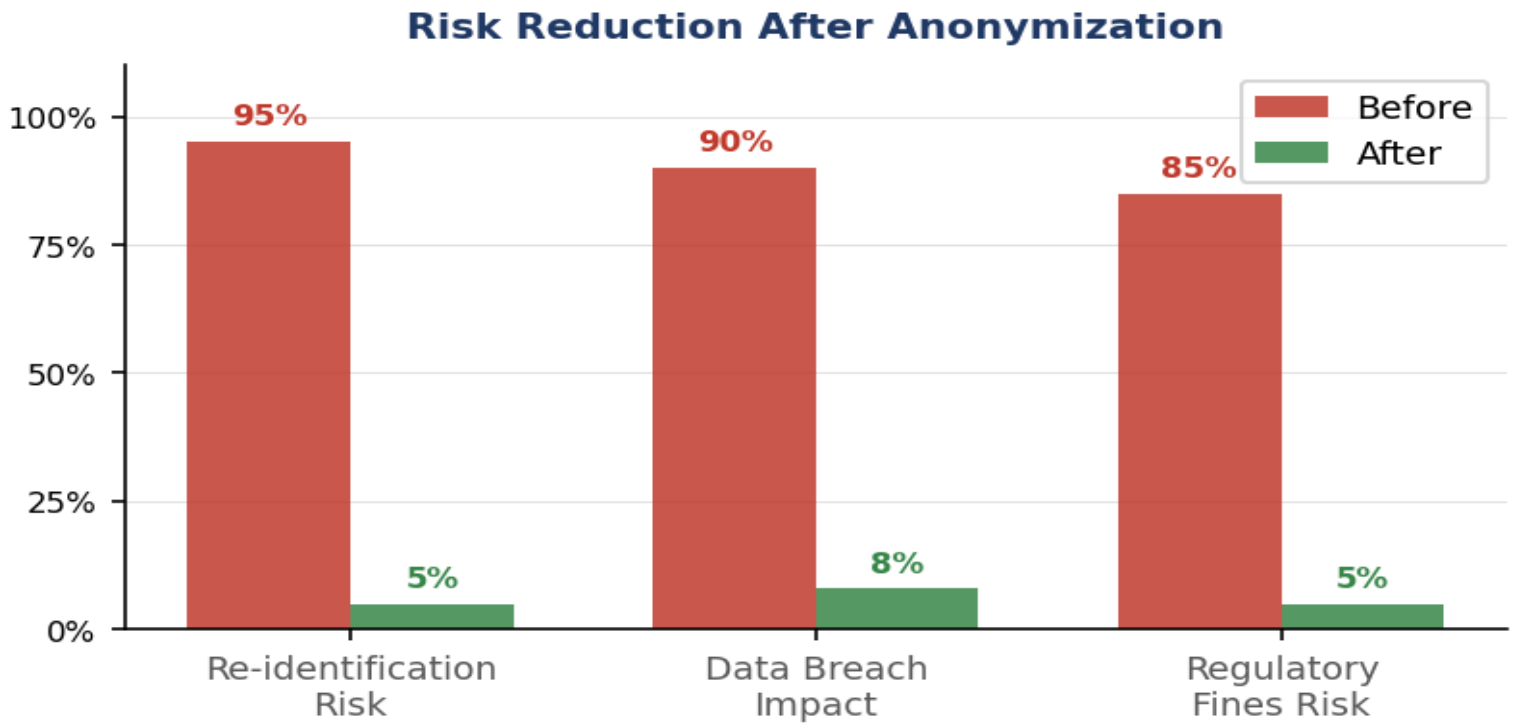
Re-identification risk, data breach impact, and regulatory fines exposure — all reduced by over 90% after anonymization.

Zero PII ever reaches the cloud

~5,000 records fully anonymized on-premises. Names, emails, addresses, IDs — none leave the local environment.

100% analytical value preserved

Country, age range, segments, categories & device type survive intact for full BI and ML. workloads



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